Guardia Cybersecurity School
Majeure Blue Team
Gestionnaire de la sécurité des données, des réseaux et des systèmes

Sécuriser l'autonomie : Cadre de défense des IA agentiques par le monitoring comportemental et les modèles souverains

Thèse de master

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Déclaration

Sauf indication contraire dans le texte ou dans les références, cette thèse est entièrement le fruit de mes propres travaux de recherche.

Lyon, France, 30 juillet 2026

. . . .
Clément RONDEAU

Résumé

Les IA agentiques se multiplient pour répondre à différents besoins au sein des entreprises. Aujourd'hui, on leur confie l'accès à des documents sensibles ainsi que la prise de décisions autonomes, ce qui en fait de nouvelles cibles critiques. Leur défense devient donc un sujet primordial pour limiter la surface d'attaque sur le système d'information. Ce mémoire explore la sécurisation de ces agents à travers le déploiement d'un monitoring comportemental en temps réel pour détecter les anomalies au runtime, telles que les injections de prompt. De plus, il démontre comment le choix de modèles souverains et frugaux, alignés sur la démarche "IA Frugale" de l'AFNOR, permet de réduire la dépendance aux API tierces, d'empêcher les fuites de données et de garantir un périmètre auditable. Cette approche est complétée par une preuve de concept (PoC) démontrant l'efficacité de ce cadre de durcissement.

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Chapitre 1

Introduction

A thesis in computer science follows the structure of academic writing and consists of several chapters, each chapter contains several sections, and each section can contain several subsections. Sections and subsections contain several paragraphs of text as well as, for example, lists, tables, and figures. It is recommended to start your thesis by studying the scientific toolbox [1].

1.1 References

Elements like sections, figures, or tables can be referenced (see Section 1.1) by assigning a `\label{label-name}` after the `\section` command or within a `\begin{...}` environment and referencing the label with `\ref{label-name}`.

1.2 Links and Citations

In scientific work, all external sources must be cited or linked.

Scientific Articles Papers, Textbooks, theses, or other scientific work should always be cited as `@article`, `@inproceedings`, or `@book` via biblatex. To cite a work, add a corresponding bibliography entry to your `MasterThesisGuardia.bib` file and reference it in the text.

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1. Find our notes on scientific work at `https://webis.de/lecturenotes.html#part-scientific-toolbox`

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Other Sources Other secondary assets, like standalone images, external libraries, or specific software source code deployments, can be dynamically referenced by providing an explicit link and a date of last access within a footnote on the current page.

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